AFR-Net, JIPNet and HERS for billion-scale fingerprint deduplication

(a) Applicability to billion-scale gallery development: design choices, latency and failure cases, with a proposed pipeline combining the three. **(b)** JIPNet and an AFR-Net-style baseline on FVC2002/2004 partial prints: EER, TAR at FAR 0.1%, JIPNet pose error and failure analysis.

Sources of figures: *reported* in the cited paper; *measured* in this study (test machine described in Part (b)); *derived* by calculation from measured or reported figures.

Executive summary

(a) Applicability to a billion-scale gallery ($\approx$ 10 billion finger records). The three papers are complementary:

- **AFR-Net**: a fixed-length template computed once per print and searchable across the full gallery. Role: **candidate retrieval**.
- **JIPNet**: accurate pairwise verification with pose estimation, strongest on partial prints. Role: **candidate verification**.
- **HERS**: search over encrypted templates. Role: **template protection**.
- Combined in a staged pipeline (a.4), they cover retrieval, verification and privacy.

(b) Partial-print experiment (FVC2002/2004, 160$\times$160 patches):

- **JIPNet accuracy** on the six real-sensor databases: EER 1.6%–3.2%, TAR at FAR 0.1% of 81%–88%, in line with the paper.
- **JIPNet pose error** (typical): 7–11° in rotation and 8–13 px (0.4–0.7 mm) in translation.
- **Comparison**: JIPNet clearly outperformed the AFR-Net-style fixed-length baseline on the same pairs, as also reported in the JIPNet paper. This confirms the value of pose estimation for partial prints and supports orientation normalisation before template extraction.
- **Main factors affecting accuracy**: fingerprint coverage within the patch, overlap between patches, rotation, and synthetic versus real data.

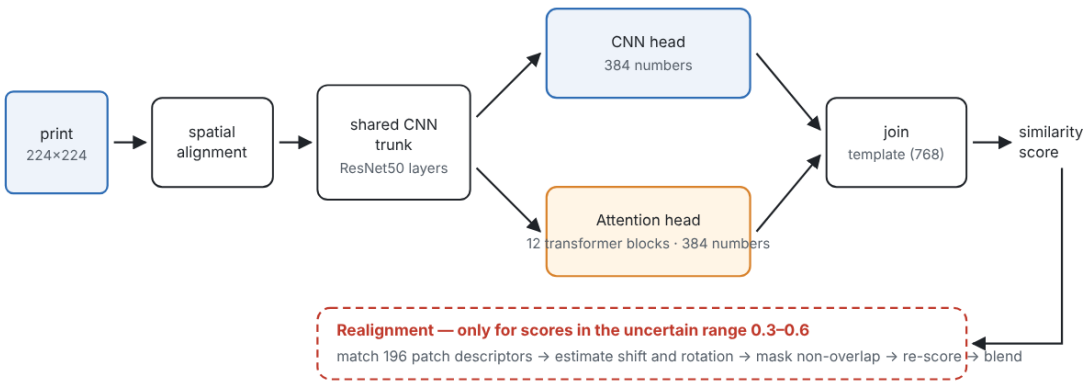
Part (a) · Applicability to billion-scale gallery development

Scale: a billion-scale gallery is taken here as 1 billion people $\times$ 10 fingers = 10 billion finger records. Every new enrolment is compared against all of them; all derived figures below scale in proportion to the gallery size.

Implication: search time and the expected number of false matches both grow in proportion to gallery size. A billion-scale system therefore needs (1) a reusable template for fast full-gallery retrieval, (2) a more accurate verifier applied to a short candidate list, and (3) fusion across fingers.

Notation used in the latency and complexity entries: N = number of finger records in the gallery ($\approx$ 10 billion); d = numbers per template; K = number of candidates passed to verification; P = local descriptors per print in AFR-Net realignment; n = templates packed into one encrypted block (ciphertext) in HERS; f = fraction of the gallery scanned when an index is used. $O(\cdot)$ describes how cost grows with these quantities, independent of hardware. GFLOP / TFLOP = 10^9 / 10^{12} floating-point operations (refer Appendix A0).

a.1 · AFR-Net: fixed-length fingerprint representation



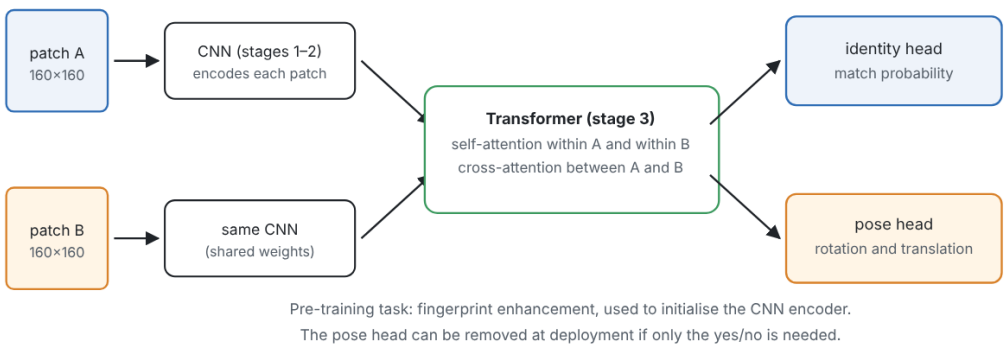
AFR-Net: shared trunk with CNN and attention heads; realignment only for uncertain scores.

Approach	**Network**: shared ResNet50 trunk with a CNN head and an attention head (85 M parameters, trained on 1.3 M images). **Template**: 384 + 384 numbers per print, compared by cosine similarity. **Realignment**: a pairwise refinement applied only to uncertain scores (0.3–0.6; 17.9% of comparisons in the authors' tests).
Reported results	**FVC2002 DB1** / **DB2** / **DB3** (full prints, standard FVC protocol, with realignment): TAR at FAR 0.1% of 100% / 99.96% / 99.36%. **NIST SD14** (rolled prints): rank-1 identification 99.78%. **NIST SD27** (latent prints vs a 100,000-print gallery): rank-1 identification 53.10%.
Design choice at scale	The template is computed once per print, stored, indexed and compatible with encryption, which makes it well suited to **candidate retrieval** over the full gallery.
Latency	**Complexity** (machine-independent): template creation $O(1)$ per print, $\approx$ 17.7 GFLOP (measured operation count); full-gallery search $O(N \cdot d) \approx$ 15 TFLOP per enrolee at d = 768 (derived); realignment $O(K \cdot P^2)$ with P = 196. **Creating a template** (once per print, at enrolment): 8.42 ms on a GPU, as reported; 1.6 s on the CPU test machine (measured). **Comparing two templates**: 233 ns on one CPU core (measured). Searching all 10 billion templates therefore takes $\approx$ 39 minutes on one core (derived), and proportionally less with more cores or an index. **Realignment** (only for uncertain scores): 29.36 ms per pair, as reported, so it is applied to candidate lists only.

Failure cases and impact

Issue	Impact at billion scale	Mitigation in the pipeline
Very low image quality (authors)	Worn, dry or wet fingers produce unreliable templates; a duplicate missed at retrieval is not recovered later.	Quality check and re-capture at enrolment (stage 0); fusion across 10 fingers.
Very small overlap (authors)	Partial or off-centre placement reduces the shared area between enrolment and query prints.	Supervised full-print capture at enrolment; JIPNet verification for partial captures (stage 2).
Rotation on small patches (refer Part (b))	Template similarity drops as rotation grows (EER 36.5% on rotated 160×160 patches; the JIPNet paper also reports 29–35% for AFR-Net on such patches).	Orientation normalisation before template extraction (stage 0).
Largest evaluated gallery 100,000 (authors)	False matches grow with gallery size; behaviour at 10 billion is not yet demonstrated.	Validation at million-scale before rollout; multi-finger fusion (stage 3).

a.2 · JIPNet: joint pairwise verification and pose estimation



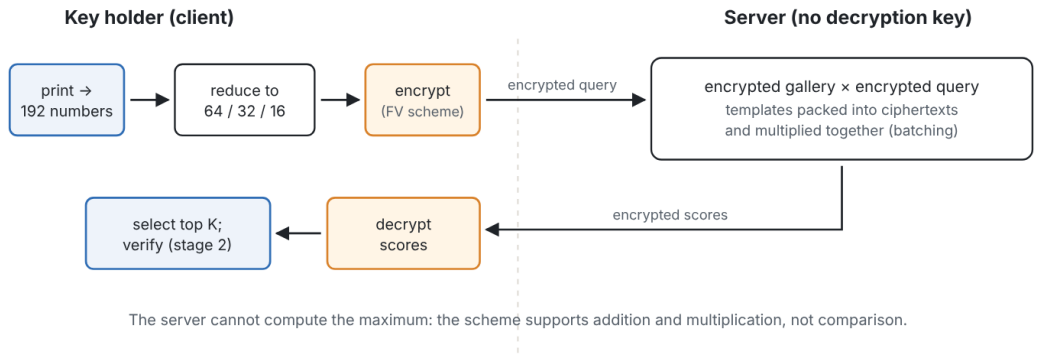
JIPNet: both patches are processed together; outputs match probability and relative pose.

Approach	**Network**: shared CNN encoder for both patches, followed by a transformer with self- and cross-attention between them. **Output**: match probability and relative pose (rotation and translation), estimated jointly.
Reported results	**Mixed partial-print test set** (160×160 patches): EER 4%, TAR at FAR 0.1% of 71% (VeriFinger: 18% and 56%). **FVC2002 DB1** (160×160 patches): EER 1%, TAR at FAR 0.1% of 93%.
Design choice at scale	A pairwise model with high accuracy on partial prints and an explicit pose output, well suited to **candidate verification** and to aligning prints for review.
Latency	**Complexity** (machine-independent): $O(K)$ network runs for K candidates, ≈ 34.6 GFLOP per pair (measured operation count); no reusable template, so a full-gallery search would be $O(N)$ runs ≈ 3.5×10^{20} FLOP (derived). **Comparing two patches**: 0.34 s per pair on the CPU test machine (measured); the paper reports latency on a GPU only as a chart (Fig. 13), without an exact value. **Verifying a candidate list** of 1,000: ≈ 6 minutes on the test machine (derived), and considerably less on a GPU.

Failure cases and impact

Issue	Impact at billion scale	Mitigation in the pipeline
Severe defects, weak texture (authors)	Low-quality enrolments may be rejected by verification even when they are true duplicates.	Fusion with other fingers (and iris/face); manual review queue for borderline cases.
Skin distortion (authors)	Pressure and skin changes between enrolments years apart; not covered by the single-capture test (refer Part (b)).	Validation on pairs from different captures before deployment.
Small overlap (authors; refer Part (b))	Pose error and missed matches increase as overlap decreases.	Minimum-coverage check at capture; several patches or fingers per decision.
Occasional 180° pose (refer Part (b))	At most 1.3% of genuine pairs, mostly with high match scores; could support a wrong alignment during review.	Pose consistency check across fingers (stage 2).
One-to-one only (authors)	Cannot search the gallery directly: 10 billion pairs would take ≈ 107 years on the test machine.	Used only after candidate retrieval (stage 1).

a.3 · HERS: search over homomorphically encrypted templates



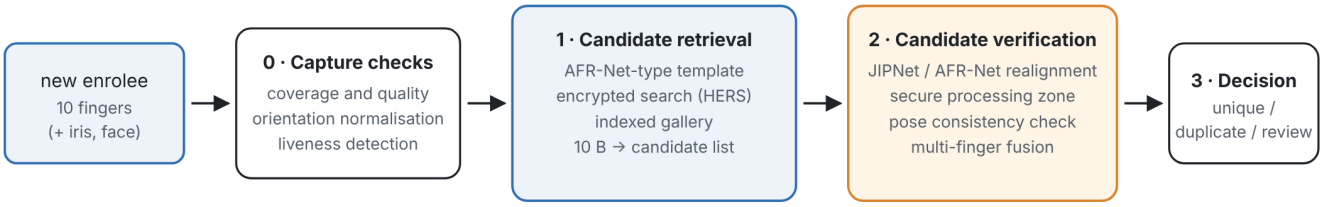
HERS: templates are reduced, encrypted and scored without decryption; only the key holder reads the scores.

Approach	**Template**: any fixed-length template, reduced to 64, 32 or 16 numbers. **Encryption**: Fan–Vercauteren homomorphic scheme (Microsoft SEAL); similarity is computed without decrypting the gallery or the query.
Reported results	**Search time** (100 million gallery, 10 CPU cores): 740 s per query at 32 numbers; ≈ 500 s with a two-stage search. **Accuracy** (NIST SD4 vs 100 million): rank-1 92.3% at 192 and 64 numbers, 92.2% at 32, 78.6% at 16. **Storage**: 140 GB at 32 numbers per 100 million templates. Reported as 275× faster than earlier encrypted search.
Design choice at scale	Templates remain protected at rest and during search, and any fixed-length template can be used, which makes it the **privacy layer** for candidate retrieval.
Latency	**Complexity** (machine-independent): $O(d \cdot \lceil N/n \rceil)$ encrypted multiplications per query, with n = 4,096 templates per ciphertext (paper setting): $\approx 7.8 \times 10^7$ at d = 32 and N = 10 billion (derived); storage $O(N \cdot d)$. **Reported**: 740 s per query for a 100 million gallery on 10 CPU cores (32 numbers per template). **Scaled to 10 billion** (× 100): ≈ 21 hours per query for a full scan (derived). With an index that limits the search to 1% of the gallery: ≈ 12 minutes (design target).

Failure cases and impact

Issue	Impact at billion scale	Mitigation in the pipeline
Accuracy drop at 16 numbers (authors)	Limits how far templates can be compressed to save time and storage.	Use 32–64 numbers per template (≈ 14 TB encrypted at 32 numbers for 10 billion).
Encrypted search cost (authors)	A full encrypted scan of 10 billion is ≈ 21 hours per query on 10 cores.	Index that narrows the search (design target 1% of the gallery, ≈ 12 minutes).
Server cannot rank results (authors)	Scores must be decrypted by the key holder to select candidates.	Ranking inside the secure processing zone, with controlled key management.
Synthetic evaluation gallery (authors; refer Part (b))	Synthetic prints behave differently from real ones (JIPNet EER 5.9%–10.0% vs 1.6%–3.2%), so real-print accuracy at scale is unverified.	Validation on a large real-print gallery.

a.4 · Proposed pipeline



Proposed staged pipeline.

Stage	Supporting evidence	Time per enrolee
0 · Capture checks	**Coverage:** JIPNet EER on FVC2002 DB1 improves from 2.5% to 1.1% when both patches are at least 70% fingerprint. **Orientation:** fixed-length templates perform best on consistently oriented prints. **Liveness:** synthetic prints score highly against each other (refer Part (b)).	Negligible
1 · Candidate retrieval	**Scalability:** fixed-length templates can be searched across the full gallery (a.1). **Privacy:** templates can stay encrypted during search (a.3). **Compactness:** 32–64 numbers per template retain accuracy (HERS).	Unencrypted: ≈ 39 minutes on one core for a full scan, reduced by more cores and indexing. Encrypted: ≈ 12 minutes on 10 cores if an index limits the search to 1% (derived).
2 · Candidate verification	**Accuracy on partial prints:** on the same pairs, JIPNet EER 2.2%–3.0% versus 36.5%–37.5% for a fixed-length template (refer Part (b)). **Pose output:** supports alignment checks and examiner review.	For 1,000 candidates: ≈ 29 s with AFR-Net realignment; ≈ 6 minutes with JIPNet on the CPU test machine.
3 · Decision	**Multi-finger fusion:** combining 10 fingers adds independent evidence and reduces false matches at gallery scale.	Negligible

Machine-independent cost per new enrolee

Step	Complexity	Operations at N = 10 billion
Template creation (AFR-Net)	$O(1)$ per print	≈ 17.7 GFLOP per print, × 10 fingers
Full-gallery search, plain	$O(N \cdot d)$	≈ 15 TFLOP at d = 768; ≈ 0.6 TFLOP at d = 32
Indexed search (fraction f)	$O(f \cdot N \cdot d)$	f = 1%: ≈ 6.4 GFLOP at d = 32
Full-gallery search, encrypted (HERS)	$O(d \cdot \lceil N/n \rceil)$ ciphertext mult.	$\approx 7.8 \times 10^7$ at d = 32, n = 4,096
Candidate verification (JIPNet)	$O(K)$	K = 1,000: ≈ 34.6 TFLOP
Decision (fusion)	$O(K \cdot \text{fingers})$	negligible

Network counts measured with PyTorch's operation counter (AFR-Net: the JIPNet authors' re-implementation at 224×224 input). These counts are independent of hardware; the wall-clock times above are one example on the test machine. Verifying 1,000 candidates with JIPNet requires more operations than a plain scan of the entire gallery at d = 32, which is why the candidate list must remain short.

Operating modes

- **Full-print enrolment** (supervised capture of full impressions): AFR-Net reports TAR at FAR 0.1% of 99.36–100% on full FVC2002 prints, where its built-in alignment can orient the print. An AFR-Net-type template can therefore handle both retrieval and verification, entirely in encrypted form.
- **Partial or low-quality captures** (small sensors, poor placement): JIPNet verifies the candidate list inside the secure processing zone, where its pose estimation gives the largest benefit (refer Part (b), section b.2).

	AFR-Net-type template	JIPNet
Stored per finger	768 numbers ≈ 3 KB (float32)	Fingerprint image (e.g. 388×374 px ≈ 145 KB uncompressed)
Full gallery, 10 B fingers	≈ 31 TB unencrypted; ≈ 14 TB encrypted at 32 numbers (derived)	Image archive for all fingers
Matching in encrypted form	Supported (HERS)	Requires decrypted images, within the secure zone

Part (b) · JIPNet and AFR-Net-style baseline on FVC2002/2004

- **Data:** FVC2002 and FVC2004, 8 databases (6 real sensors, 2 synthetic), each with 100 fingers × 8 impressions.
- **Test pairs:** 160×160 patches, following the JIPNet protocol (patch centres up to 100 px apart, rotation between $-180°$ and $180°$). Per database: 800 genuine pairs (same finger) and 2,400 impostor pairs (different fingers).
- **Ground-truth pose:** both patches of a genuine pair are cut from one impression with a known shift and rotation, so the true pose is exact. Results are therefore an upper bound for pairs taken from different captures.
- **JIPNet:** official code and weights; outputs reproduced to 2 decimals on the authors' examples. Run on all pairs of all 8 databases.
- **AFR-Net-style baseline:** AFR-Net weights released by the JIPNet authors, used as a fixed-length template (score = 0.2 ×

CNN similarity + 0.8 × attention similarity). Run on a subsample: 400 genuine pairs each from FVC2002 DB1 and DB3, against all different-finger combinations.

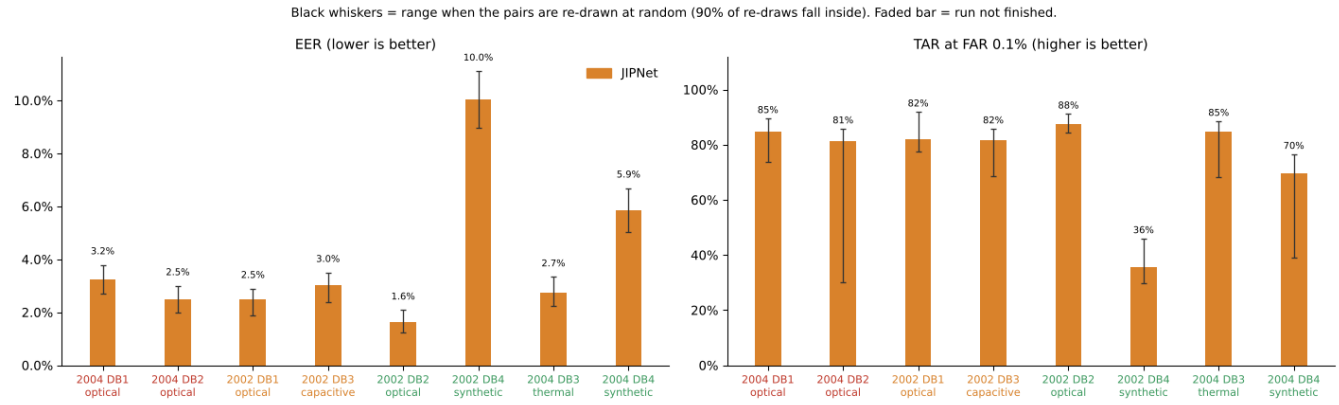
- **Metrics:** *EER*, the error rate at the threshold where false accepts equal false rejects (lower is better). *TAR at FAR 0.1%*, the share of genuine pairs accepted when only 1 in 1,000 impostor pairs is accepted (higher is better). *Pose error*, the difference between estimated and true rotation (degrees) and translation (pixels).
- **Test machine:** 2-core Intel Xeon CPU (2.1 GHz), no GPU. Measured timings refer to this machine; a GPU server is considerably faster.

b.1 · EER and TAR at FAR 0.1% (JIPNet)

Database	EER	TAR at FAR 0.1%
FVC2002 DB1 (optical) paper's test set	**2.5%** (1.9%–2.9%)	**82%** (78%–92%)
FVC2002 DB2 (optical)	**1.6%** (1.2%–2.1%)	**88%** (84%–91%)
FVC2002 DB3 (capacitive) paper's test set	**3.0%** (2.4%–3.5%)	**82%** (68%–86%)
FVC2002 DB4 (synthetic)	**10.0%** (9.0%–11.1%)	**36%** (30%–46%)
FVC2004 DB1 (optical) JIPNet trained on it	**3.2%** (2.7%–3.8%)	**85%** (74%–90%)
FVC2004 DB2 (optical) JIPNet trained on it	**2.5%** (2.0%–3.0%)	**81%** (30%–86%)
FVC2004 DB3 (thermal)	**2.7%** (2.2%–3.3%)	**85%** (68%–88%)
FVC2004 DB4 (synthetic)	**5.9%** (5.0%–6.7%)	**70%** (39%–77%)

Values in brackets: range across random re-draws of the test pairs (90% interval).

Comparison with the paper (FVC2002 DB1: EER 1%, TAR 93%; DB3: EER 3%, TAR 67%). TAR at FAR 0.1% is set by the few highest-scoring impostor pairs, so it varies widely between samples (78–92% here on DB1). The remaining difference is explained by blank area in the patches: when both patches are at least 70% fingerprint, JIPNet reaches EER 1.1% (paper: 1%) and TAR 89% (paper: 93%) on FVC2002 DB1. On DB3 the measured TAR (82%) is higher than the paper's (67%).



EER and TAR at FAR 0.1% per database.

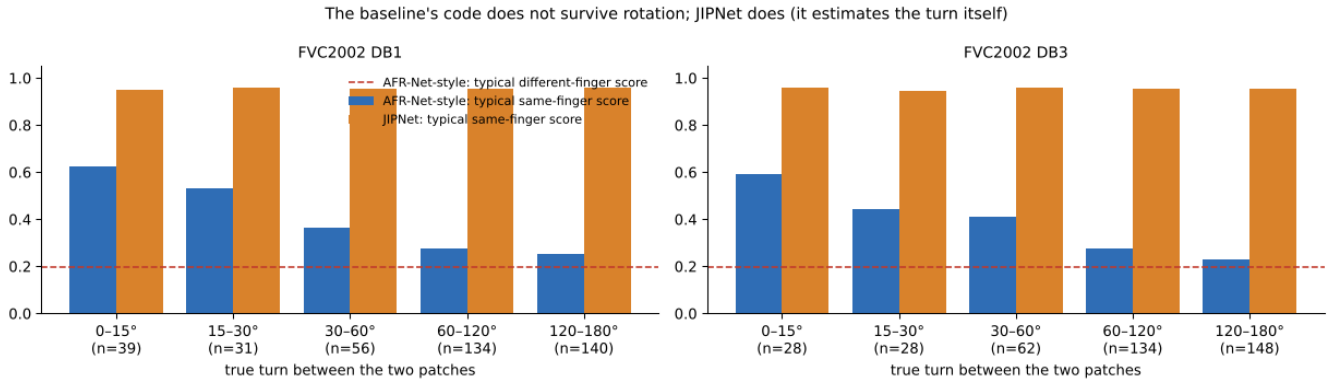
b.2 · Comparison with the AFR-Net-style baseline

Database	JIPNet, measured	JIPNet, paper	AFR-Net-style, measured	AFR-Net, paper*
FVC2002 DB1 (optical)	**2.2%** / 82%	1% / 93%	**36.5%** / 2%	29% / 3%
FVC2002 DB3 (capacitive)	**3.0%** / 81%	3% / 67%	**37.5%** / 2%	35% / 1%

Each cell: EER / TAR at FAR 0.1%. Measured values use the same 400 genuine pairs per database for both models. Paper values: JIPNet paper, Table V, same patch size, shift and rotation ranges (the paper cuts the two patches from different impressions). *The paper's AFR-Net result adds a separate alignment network in front of AFR-Net.

- **Agreement with the paper:** both measured models are close to the values reported in the JIPNet paper. The paper likewise reports a high EER for AFR-Net on these partial patches (29% on DB1, 35% on DB3), even with an added alignment network; AFR-Net's near-perfect published results refer to full prints.
- **Effect of rotation:** the fixed-length template is not rotation-invariant. On FVC2002 DB1 its typical genuine score is 0.62 for

rotations under 15° and 0.25 for rotations of 120–180°. JIPNet estimates the rotation itself and stays near 0.95 at all angles.



Typical genuine score by true rotation between the patches.

b.3 · JIPNet pose error

Database	Rotation, typical	Rotation, worst 10%	Translation, typical	Translation, worst 10%
FVC2002 DB1 (optical)	9.1°	24°	8.6 px (0.44 mm)	20 px
FVC2002 DB2 (optical)	7.9°	23°	8.7 px (0.44 mm)	21 px
FVC2002 DB3 (capacitive)	7.3°	22°	8.3 px (0.42 mm)	24 px
FVC2002 DB4 (synthetic)	11.4°	43°	13.4 px (0.68 mm)	44 px
FVC2004 DB1 (optical)	8.6°	25°	9.4 px (0.48 mm)	26 px
FVC2004 DB2 (optical)	7.7°	24°	8.7 px (0.44 mm)	24 px
FVC2004 DB3 (thermal)	8.4°	28°	9.7 px (0.49 mm)	27 px
FVC2004 DB4 (synthetic)	10.7°	36°	12.5 px (0.64 mm)	37 px

Genuine pairs only. Typical = median (half the pairs have a smaller error). Worst 10% = the error exceeded by only 1 pair in 10. 1 px = 0.0508 mm at 500 dpi. Pose error grows as overlap decreases (Figure A4).

b.4 · Failure analysis

Factor	Observation and mitigation
Fingerprint coverage	JIPNet EER on FVC2002 DB1 improves from 2.5% to 1.1% (paper: 1%) when both patches are $\geq$ 70% fingerprint (Figure A5). Lowest-scoring genuine pairs often contain blank regions (Figure A6).
Overlap	Rotation and translation errors increase as the overlap between patches decreases (Figure A4).
Synthetic prints	EER 5.9%–10.0% versus 1.6%–3.2% on real sensors; different synthetic fingers receive high scores (Figure A7). This suggests synthetic galleries are not representative of real performance, and that resistance to synthetic prints (e.g. “MasterPrint” attacks, Roy et al. 2017; Bontrager et al. 2018) should be part of evaluation.
180° pose estimates	Rotation error above 150° in at most 1.3% of genuine pairs (0–0.3% on real sensors), mostly with high match scores; a pose consistency check addresses this.
Rotation (fixed-length)	Accuracy of the AFR-Net-style template decreases with rotation (refer Part (b), section b.2); orientation normalisation before template extraction addresses this.
Sensor type	Consistent performance on real sensors not seen in training (optical FVC2002 DB2, thermal FVC2004 DB3).

Appendix

A0 · Note on operation counts (GFLOP). A *FLOP* is one floating-point operation: a single addition or multiplication of two decimal numbers. A *GFLOP* is one billion such operations. For example, creating one AFR-Net template takes ≈ 17.7 GFLOP, and comparing one JIPNet pair takes ≈ 34.6 GFLOP. The operation count is fixed by the method and does not depend on the hardware, which is why it is used here as a machine-independent measure of cost. Hardware speed is expressed separately in GFLOPS (operations per second). The time for a step is approximately the operation count divided by the hardware speed, so the same count gives shorter times on faster hardware such as a GPU.

Supporting figures

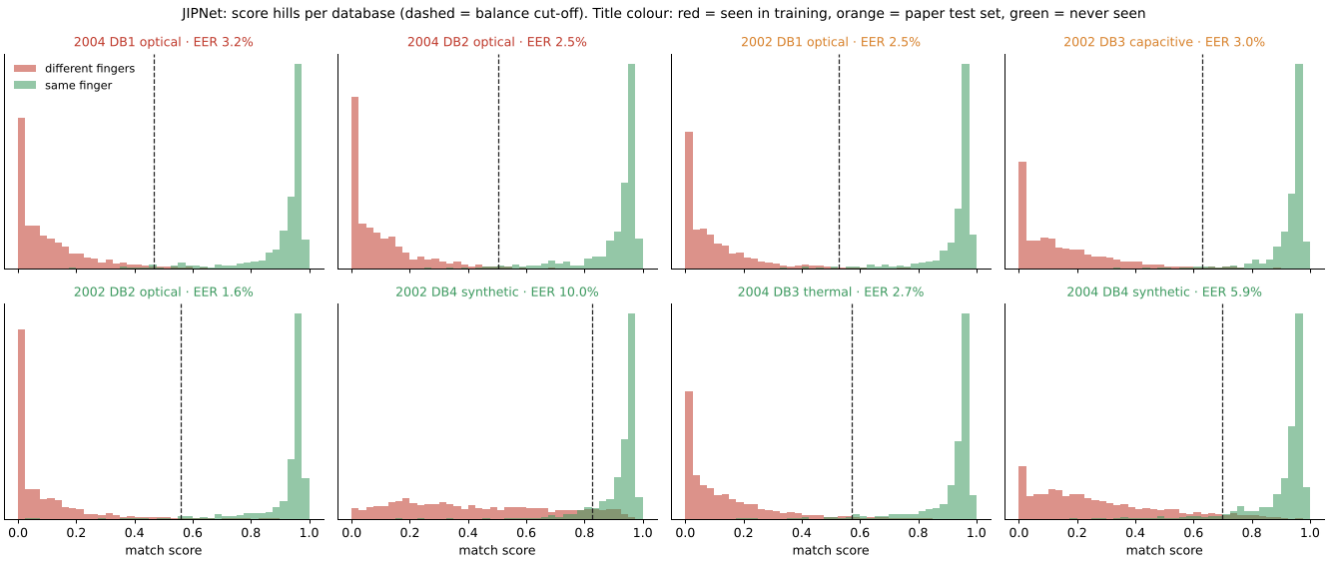


Figure A1. JIPNet score distributions per database: same-finger (green) and different-finger (red) pairs.

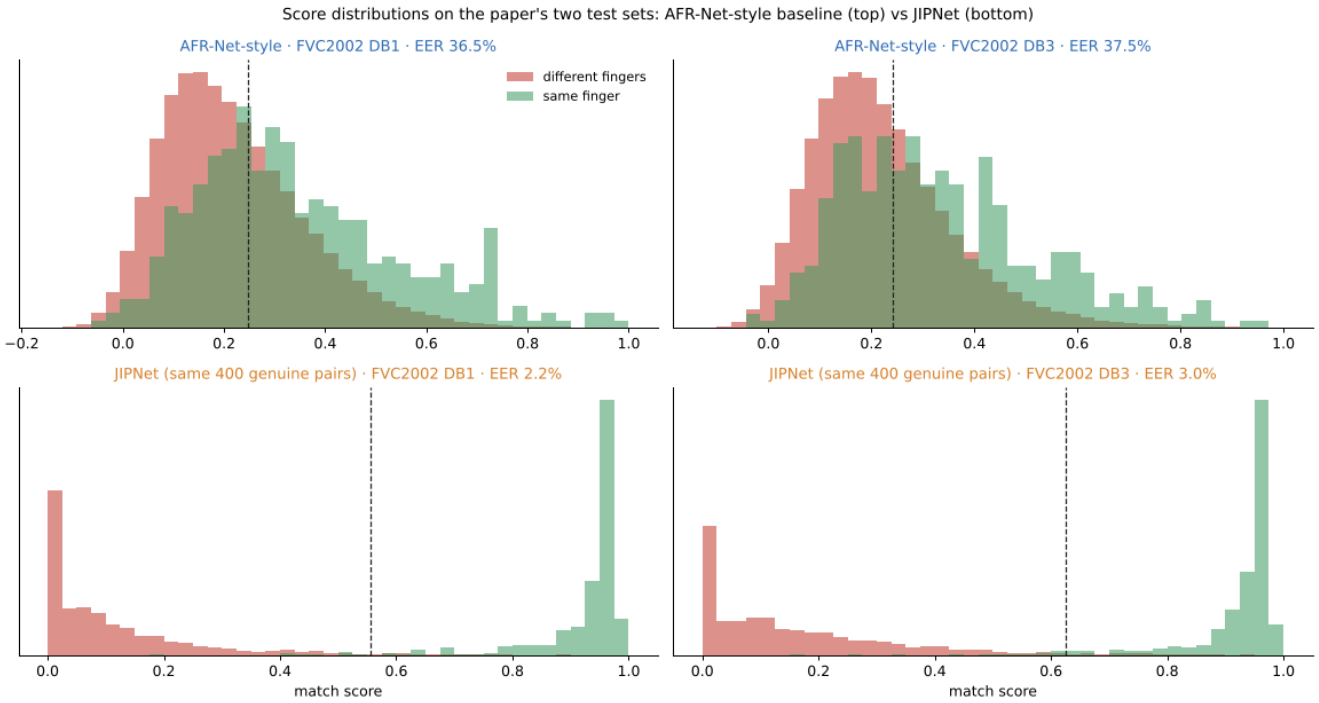


Figure A2. Score distributions on FVC2002 DB1/DB3: AFR-Net-style baseline (top) and JIPNet (bottom).

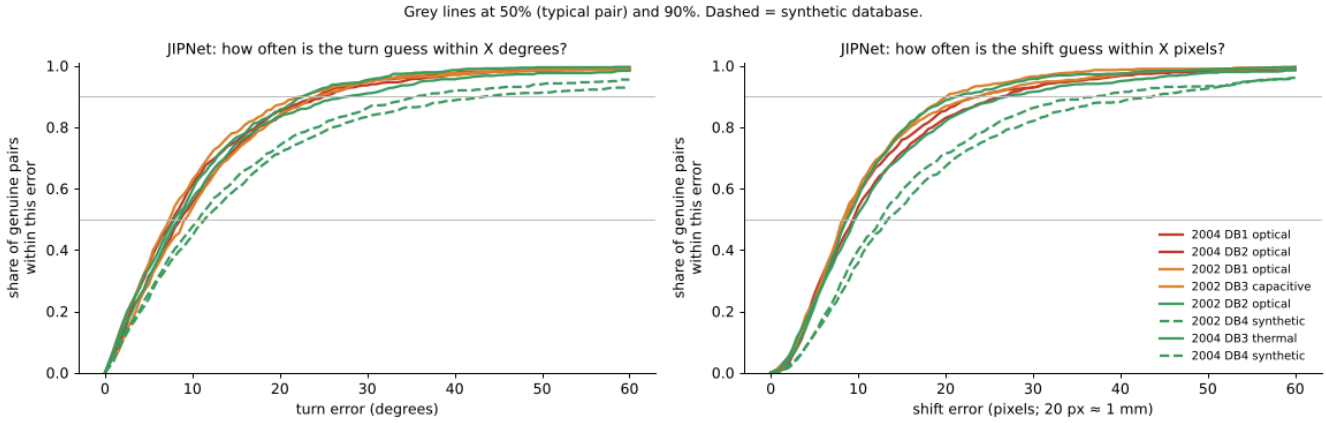


Figure A3. JIPNet: share of genuine pairs within a given rotation or translation error.

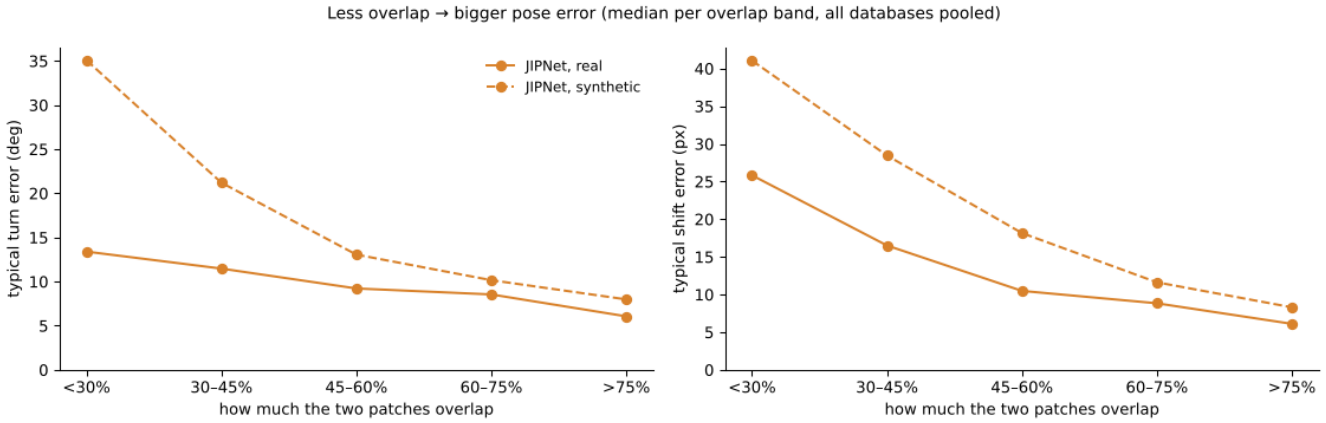


Figure A4. JIPNet: pose error by overlap between the two patches.

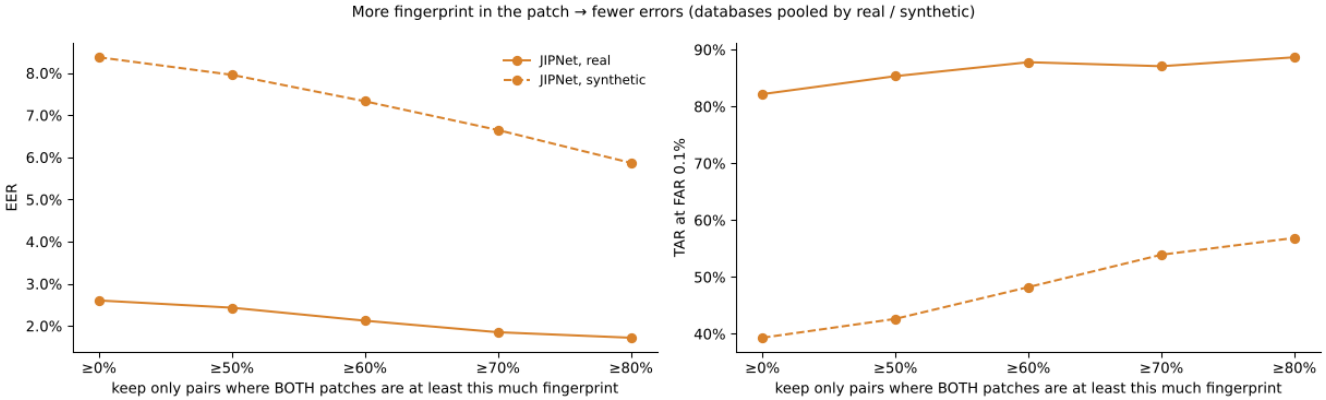


Figure A5. Effect of fingerprint coverage within the patch on EER and TAR at FAR 0.1%.

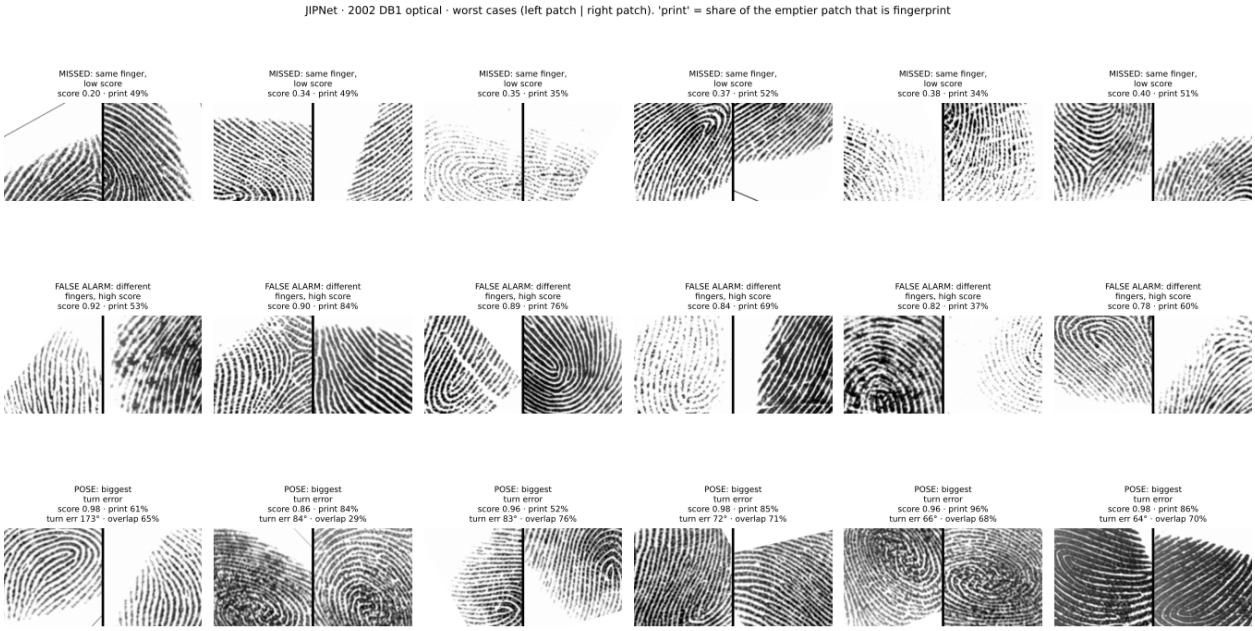


Figure A6. JIPNet lowest-scoring genuine pairs, highest-scoring impostor pairs and largest rotation errors, FVC2002 DB1.

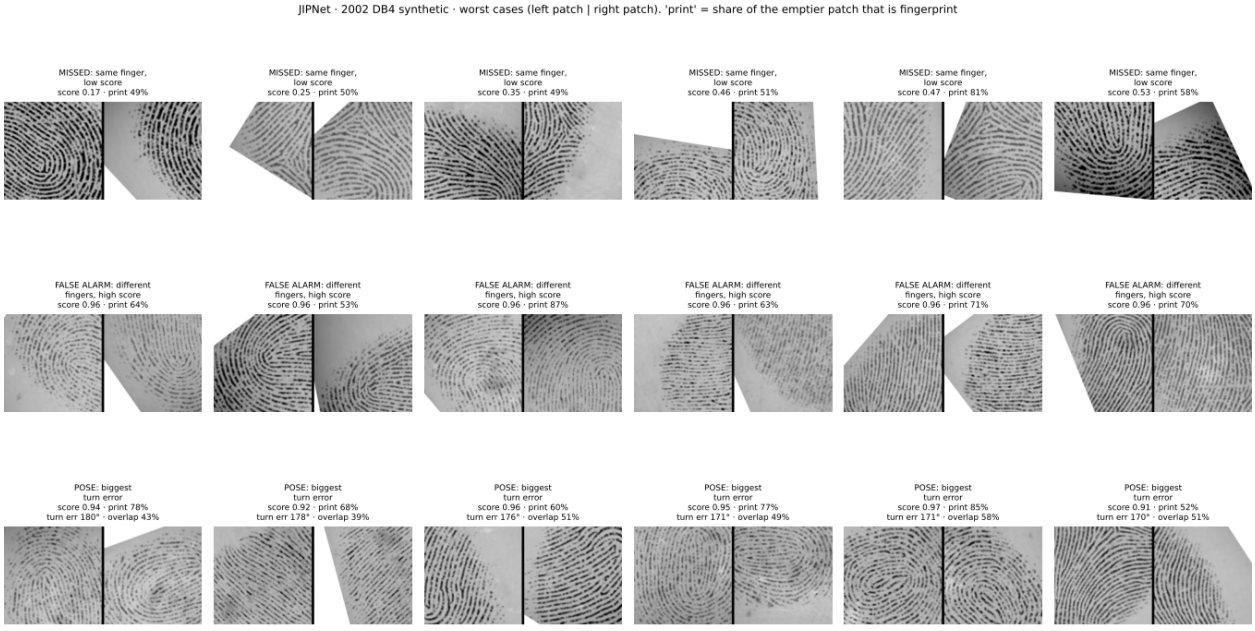


Figure A7. As Figure A6, synthetic FVC2002 DB4.

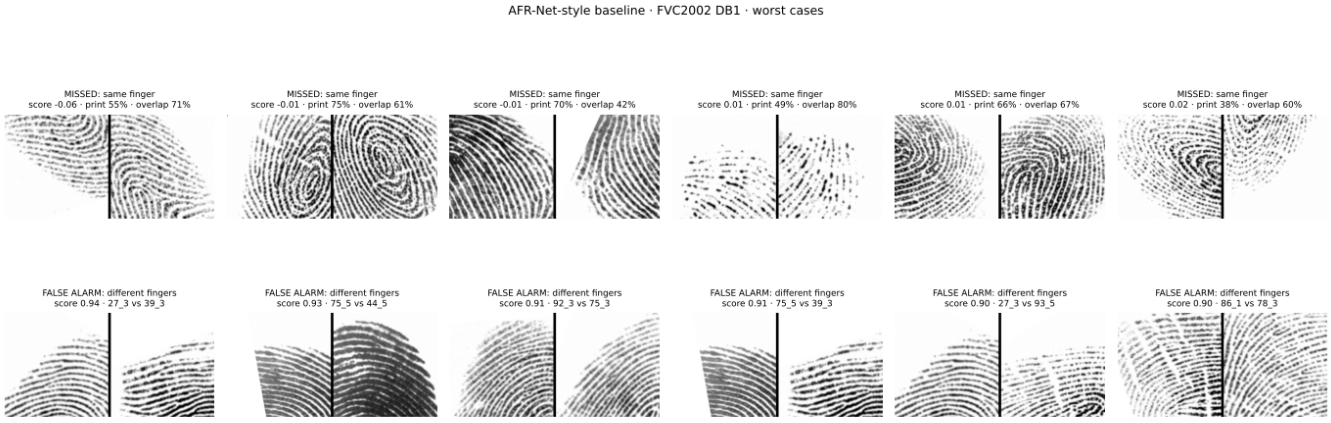


Figure A8. AFR-Net-style baseline lowest-scoring genuine and highest-scoring impostor pairs, FVC2002 DB1.

Sources: AFR-Net, arXiv 2211.13897 · JIPNet, arXiv 2405.03959; code github.com/XiongjunGuan/JIPNet · HERS, arXiv 2003.12197 · Roy, Memon, Ross, arXiv 1705.07386 · Bontrager et al., IEEE BTAS 2018 · FVC2002/2004, bias.csr.unibo.it.